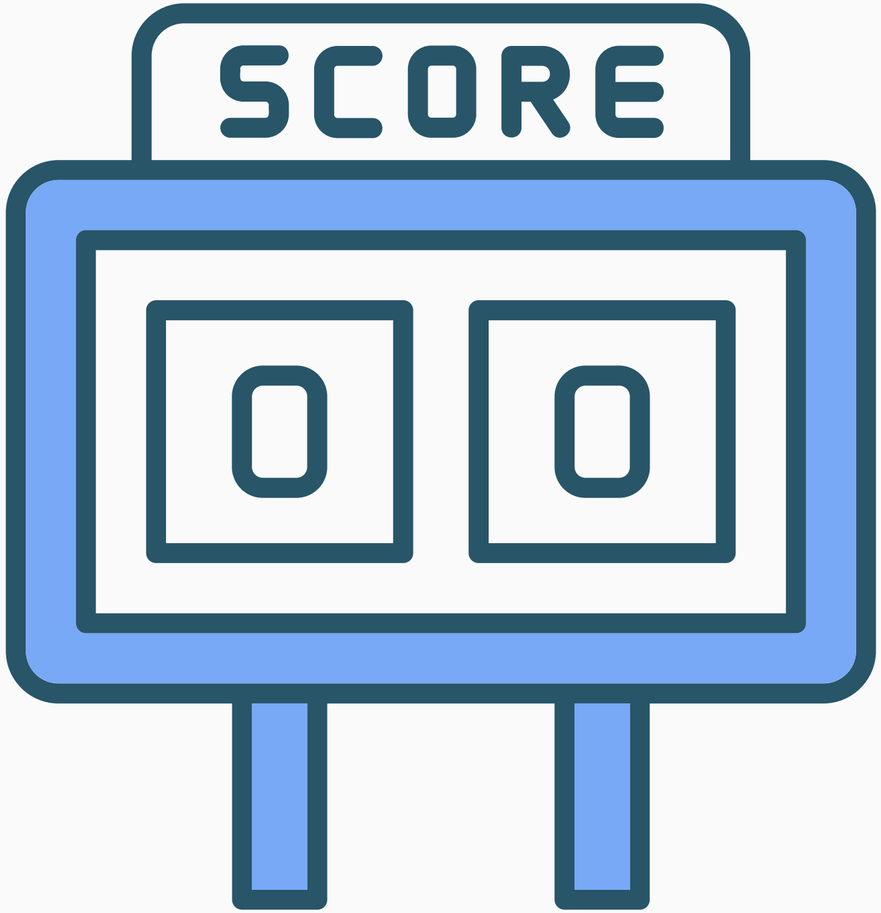


UNIVERSITÀ
DEGLI STUDI
DI MILANO

BEYOND THE SCOREBOARD.



Giuseppe Bagnara & Giovanni Passamonti

CONTEXT



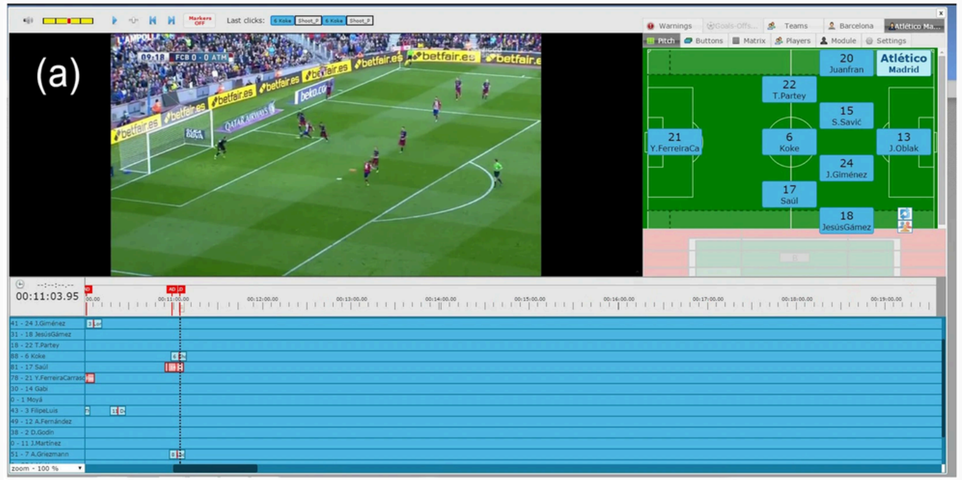
- The **2026 FIFA World Cup** hosted across 16 cities in the USA, Mexico and Canada, the first expanded edition featuring 48 teams.

- Football datasets are broadly divided into **temporal data**, tracking players' movements and positions over time, and **event & outcome data**, recording specific match actions and their results.

STATE OF THE ART

Existing open-access football datasets generally fall into two categories:

spatio-temporal event feeds



flat historical summaries



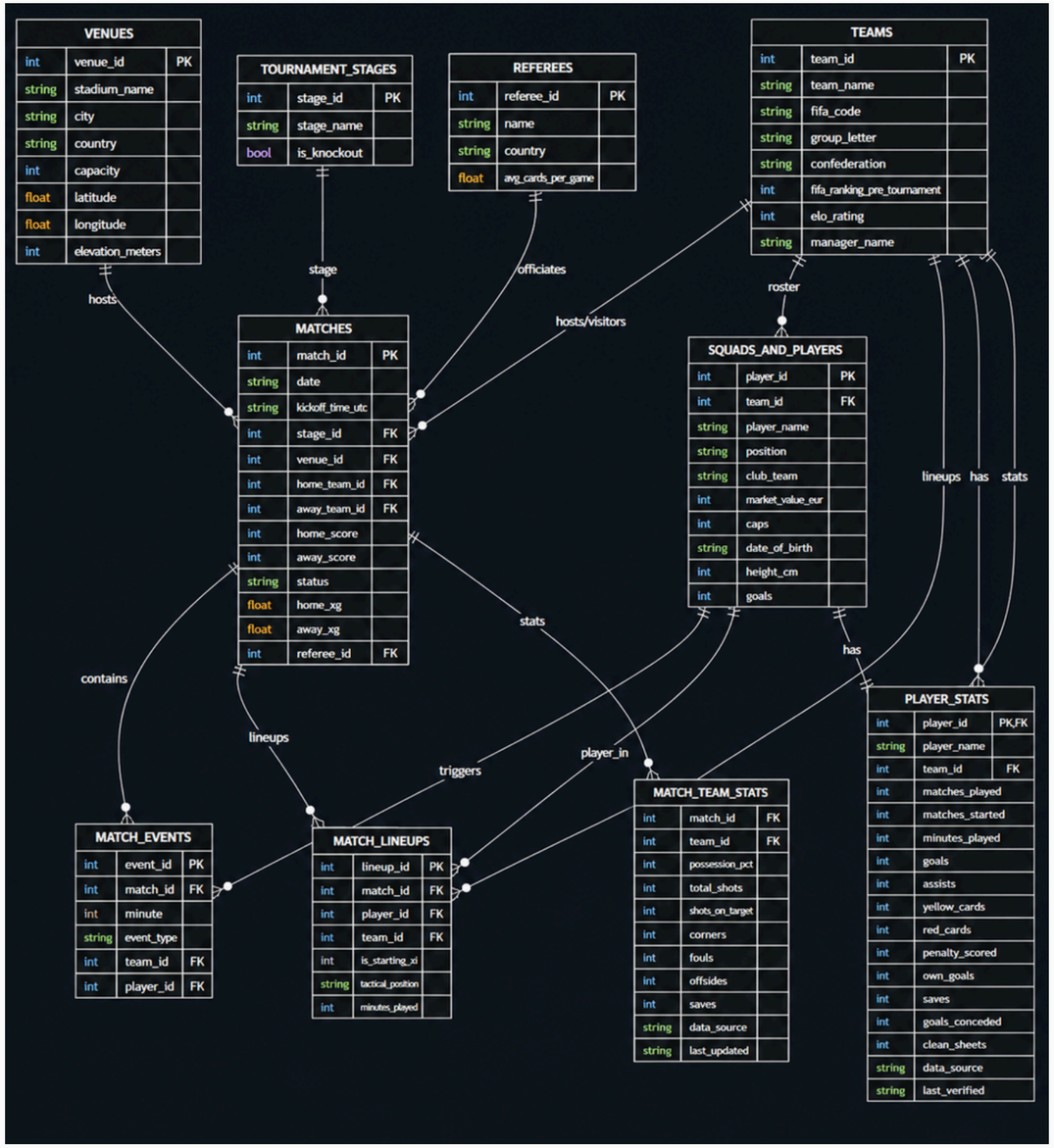
However, these datasets are **limited to historical domestic seasons** or **selective tournament subsets**, often omitting venue elevations, detailed bio-metrics, or pre-engineered machine learning features.

DATASET

- *The FIFA World Cup 2026 Master Dataset: A Normalized 3NF Relational Benchmark of the Expanded 48-Team International Football Tournament.* (MD Mominul Islam, 2026)

- Combines data from **Official FIFA bullettins**, **SofaScore Summaries** and **Transfermarkt Bio-Metrics**.

- Detailed collection of data for the **104 matches**.

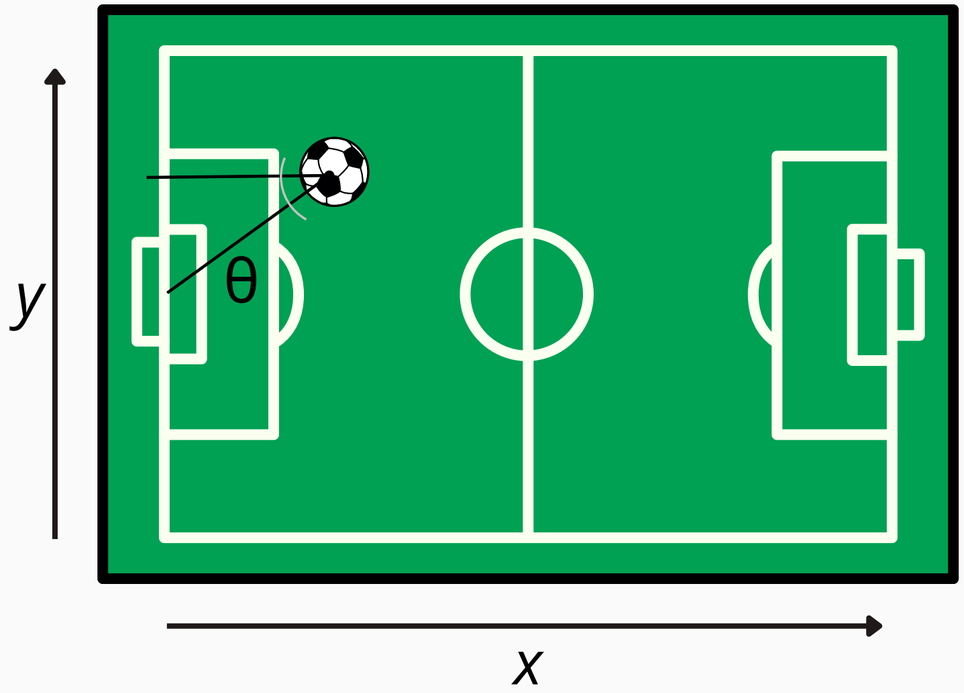


HYPOTHESIS 1

Do xG, ball possession and shots describe the result?

$$xG = \sum_{i=1}^{S} E[Goal_i | x_i, y_i, \theta_i]$$

S= shots
i= nº of shot
E= expected value
Goal= causal variable for a shot
x, y= spatial coords [0,100]
x= long side, y=short side
θ= distance & angle from the center of the goal



HYPOTHESIS 2

Are there any unexpected results?

We'll call a result **unexpected** when the difference of performance between two teams predicts a different outcome for the match.

PREPROCESSING

- The data was saved into **CSV files**
- **Some matches didn’t add up** to 100% of ball possession, so we had to **normalize**.
- For our use case, we used the *match_prediction_feature.csv* and the *match_team_stats.csv* files
- For each match, we built **two vectors with the information of the match** seen from both sides.

K-MEANS

hypothesis 1

- K = 4 classes: “Dominating”, “Equilibrated, best”, “Equilibrated, worst”, “Dominated”.
- K-means used to fit a **vector** with:
  - **xG**
  - **Ball possession**
  - **Shots data** (SOTs, total shots)
  - **Goals***.

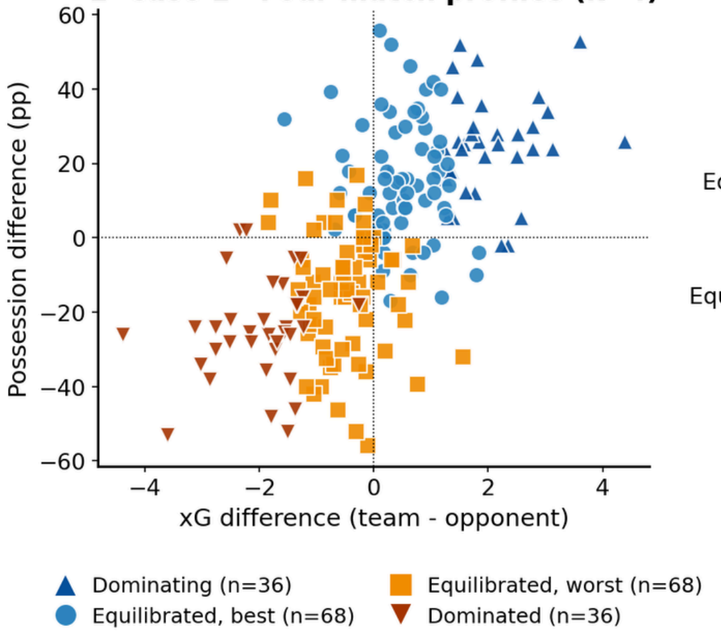
hypothesis 2

- K=2 classes: “Expected”, “Unexpected”.
- We calculate:
  - **performance index** (standard deviation of the performance)
  - **goal index** (standard deviation of the difference of goals).

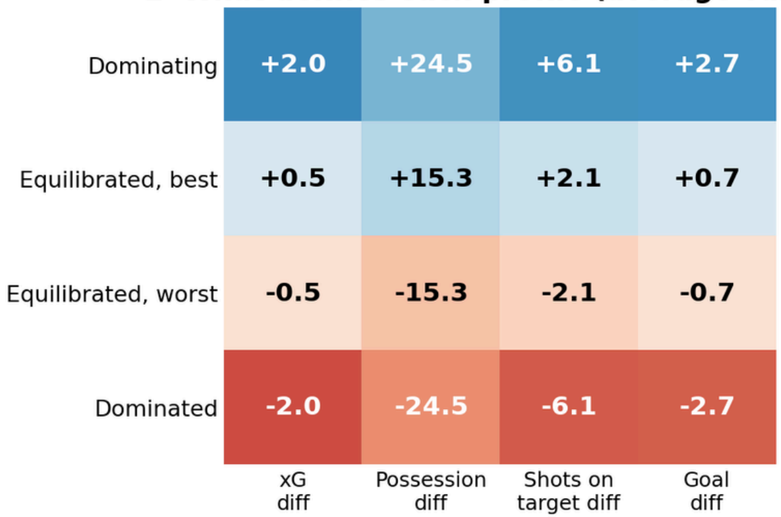
*: a different version of the hypothesis didn’t considert those as relevant

Beyond the scoreboard: do xG and possession describe the result? · FIFA World Cup 2026

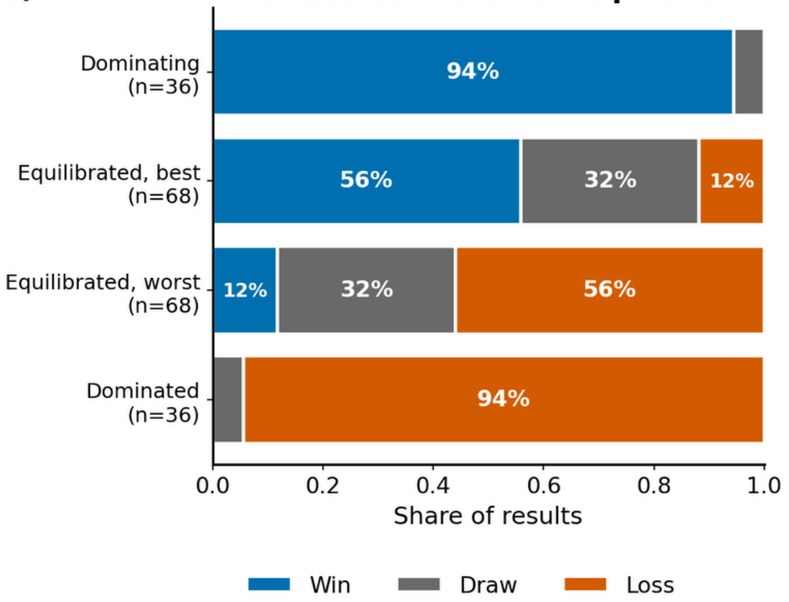
1 Case 1 · Four match profiles (k=4)



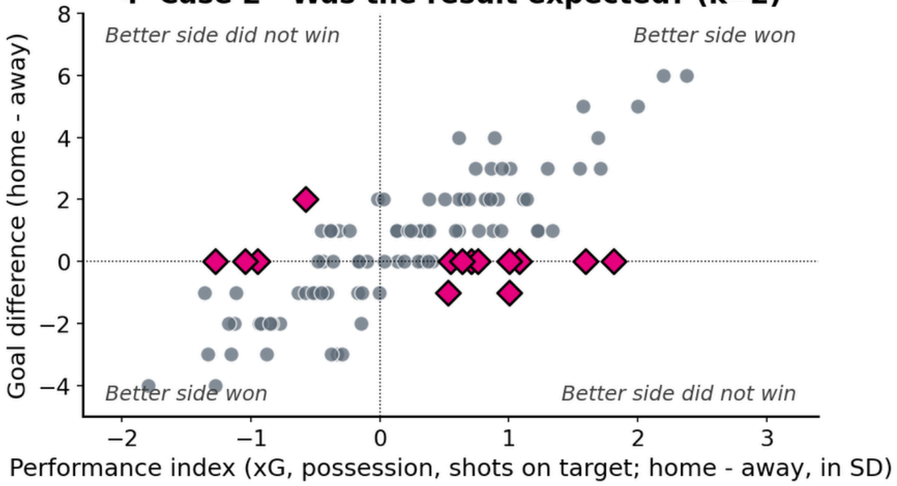
2 What defines each profile (average values)



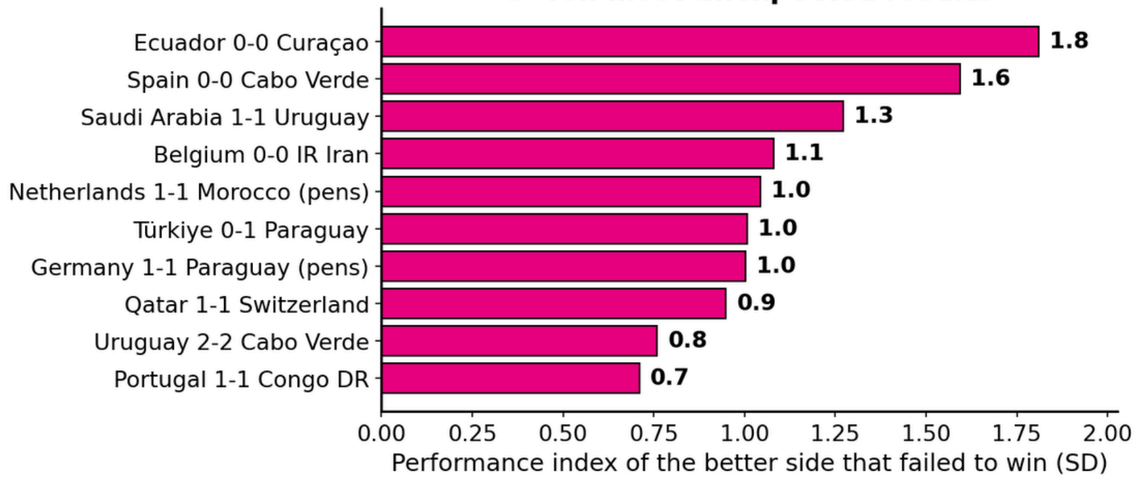
3 Results inside each profile



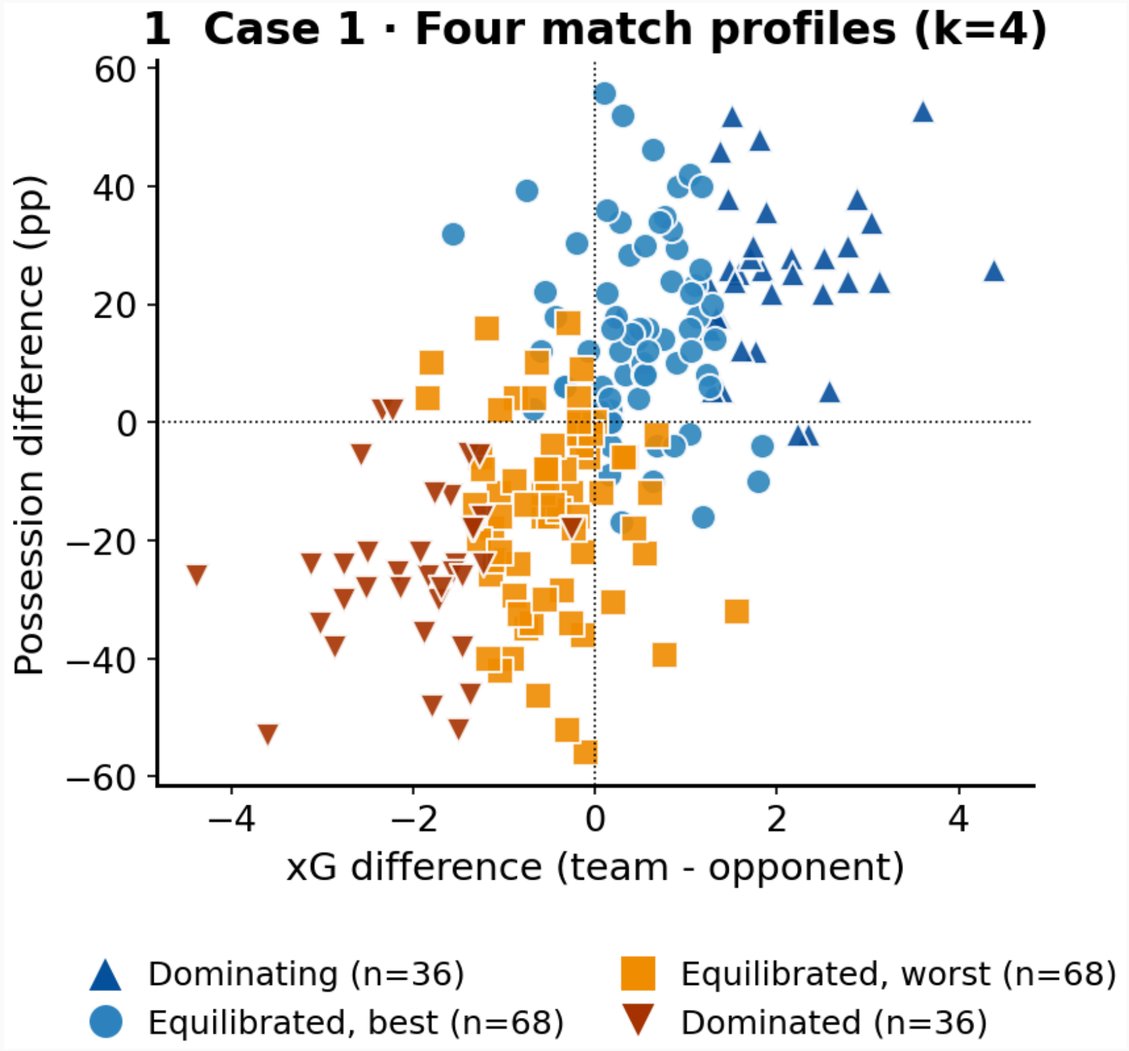
4 Case 2 · Was the result expected? (k=2)



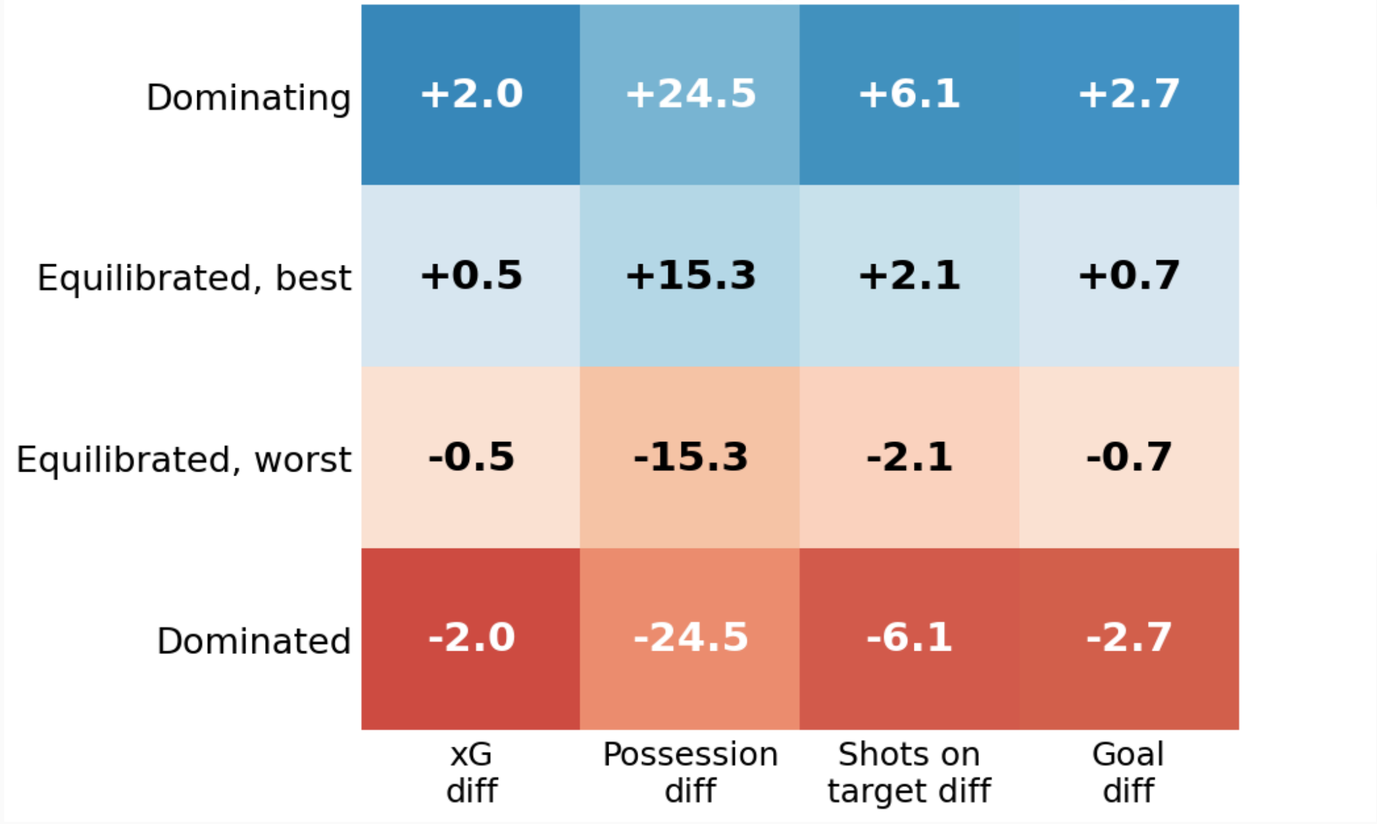
5 Ten most unexpected results

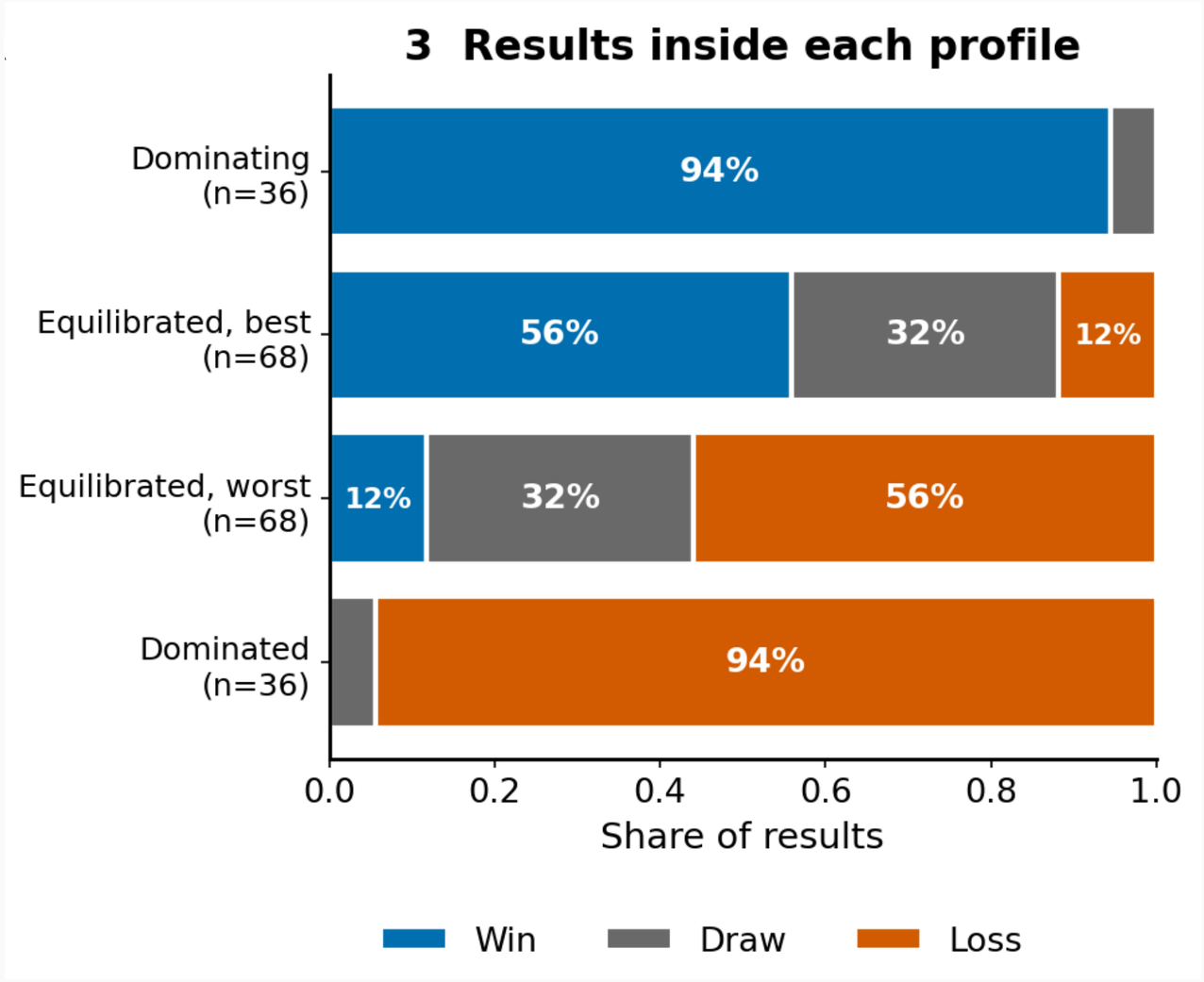


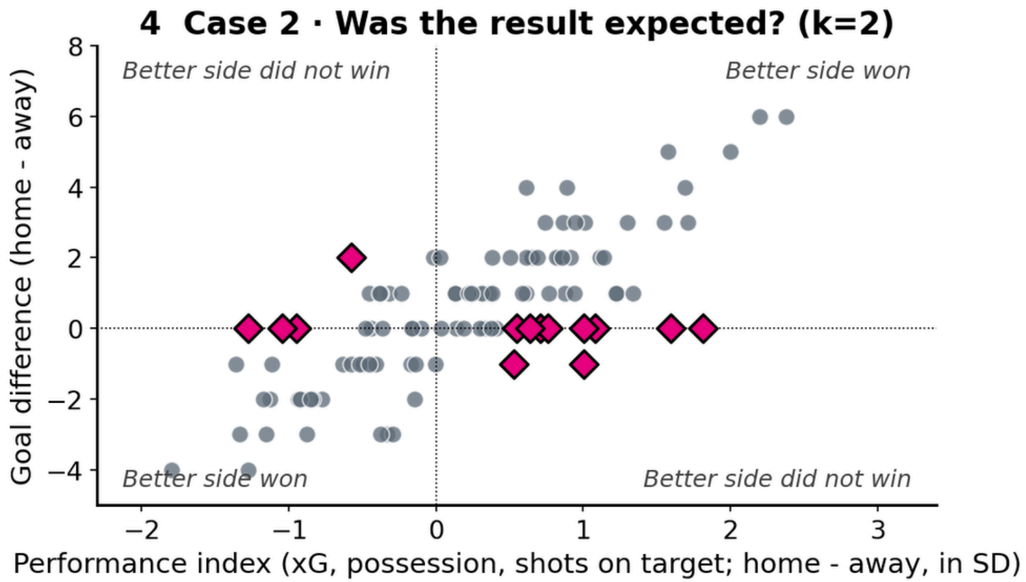
04- interpret/visualize

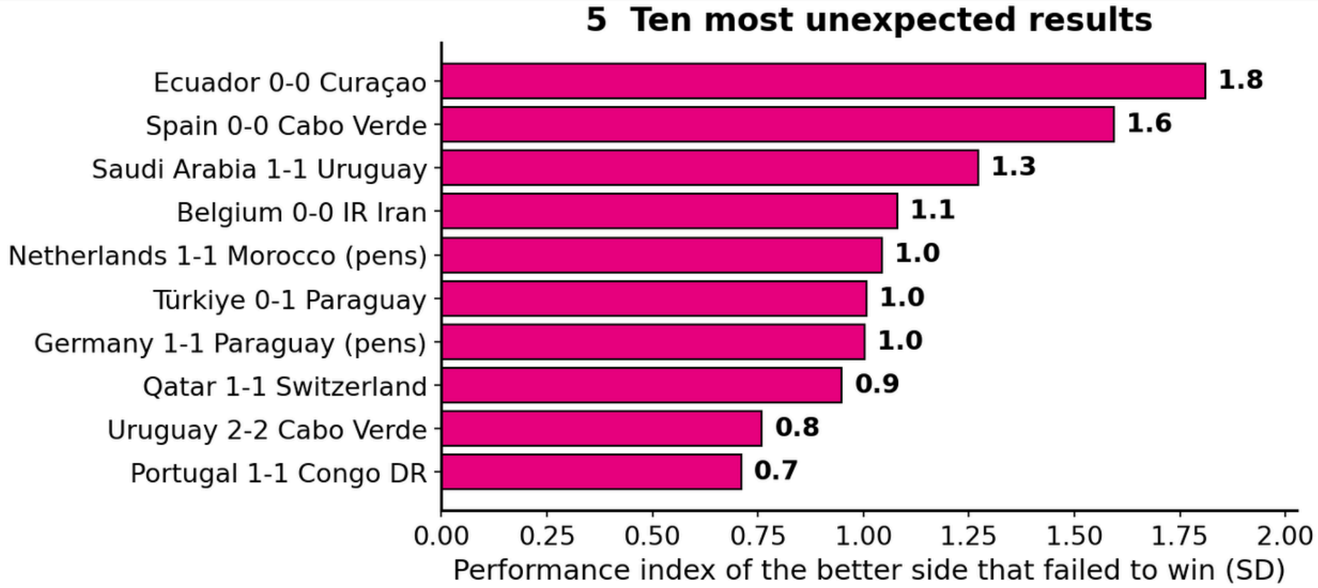

1 Case 1 · Four match profiles (k=4)
Possession difference (pp)
60
40
20
0
−20
−40
−60
−4
−2
0
2
4
xG difference (team - opponent)
Dominating (n=36)
Equilibrated, best (n=68)
Equilibrated, worst (n=68)
Dominated (n=36)

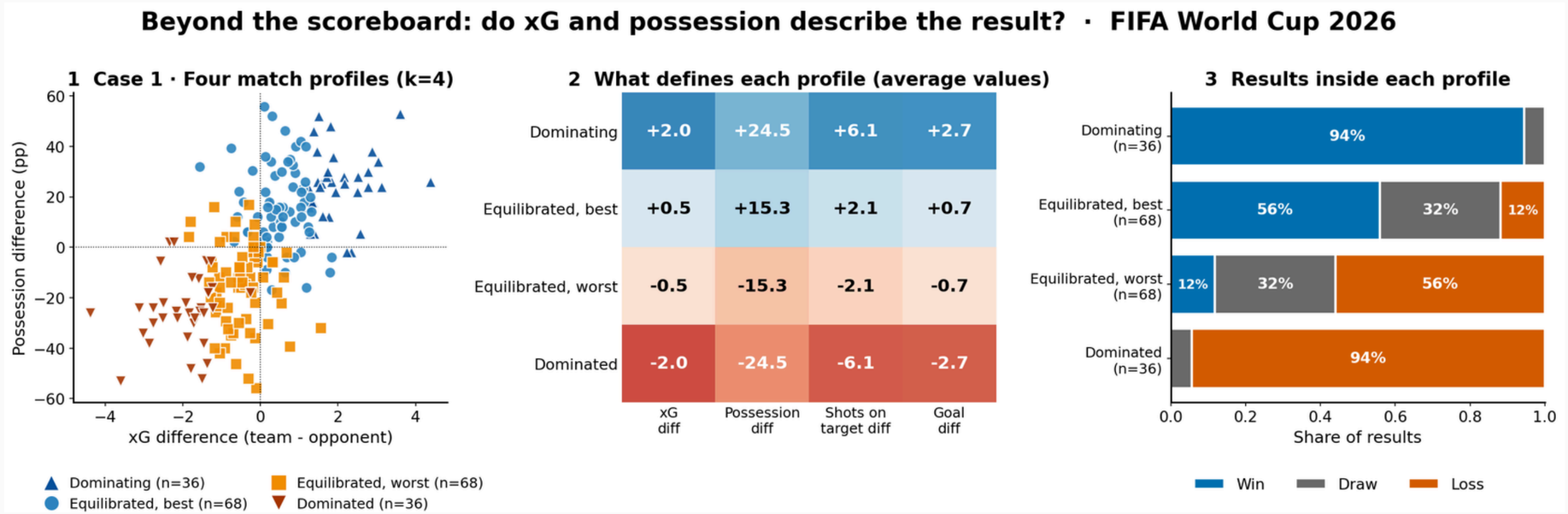
2 What defines each profile (average values)

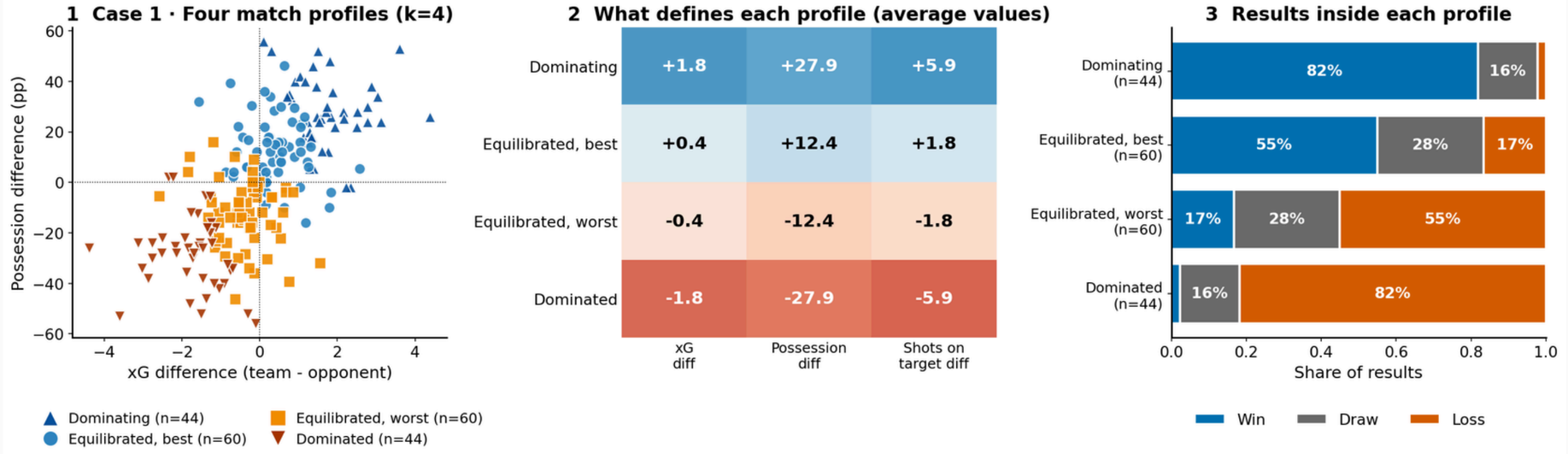



3 Results inside each profile
Dominating (n=36)
Equilibrated, best (n=68)
Equilibrated, worst (n=68)
Dominated (n=36)
94%
56%
32%
12%
12%
32%
56%
94%
0.0
0.2
0.4
0.6
0.8
1.0
Share of results
Win
Draw
Loss


4 Case 2 · Was the result expected? (k=2)
Better side did not win
Better side won
Better side won
Better side did not win
Goal difference (home - away)
8
6
4
2
0
−2
−4
−2
−1
0
1
2
3
Performance index (xG, possession, shots on target; home - away, in SD)


5 Ten most unexpected results
Ecuador 0-0 Curaçao
Spain 0-0 Cabo Verde
Saudi Arabia 1-1 Uruguay
Belgium 0-0 IR Iran
Netherlands 1-1 Morocco (pens)
Türkiye 0-1 Paraguay
Germany 1-1 Paraguay (pens)
Qatar 1-1 Switzerland
Uruguay 2-2 Cabo Verde
Portugal 1-1 Congo DR
1.8
1.6
1.3
1.1
1.0
1.0
1.0
0.9
0.8
0.7
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Performance index of the better side that failed to win (SD)


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1 Case 1 · Four match profiles (k=4)
Possession difference (pp)
xG difference (team - opponent)
Dominating (n=36)
Equilibrated, best (n=68)
Equilibrated, worst (n=68)
Dominated (n=36)
2 What defines each profile (average values)
Dominating
Equilibrated, best
Equilibrated, worst
Dominated
+2.0
+24.5
+6.1
+2.7
+0.5
+15.3
+2.1
+0.7
-0.5
-15.3
-2.1
-0.7
-2.0
-24.5
-6.1
-2.7
xG diff
Possession diff
Shots on target diff
Goal diff
3 Results inside each profile
Dominating (n=36)
Equilibrated, best (n=68)
Equilibrated, worst (n=68)
Dominated (n=36)
94%
56%
32%
12%
12%
32%
56%
94%
Share of results
Win
Draw
Loss


1 Case 1 · Four match profiles (k=4)
Possession difference (pp)
xG difference (team - opponent)
Dominating (n=44)
Equilibrated, best (n=60)
Equilibrated, worst (n=60)
Dominated (n=44)
2 What defines each profile (average values)
Dominating
Equilibrated, best
Equilibrated, worst
Dominated
+1.8
+27.9
+5.9
+0.4
+12.4
+1.8
-0.4
-12.4
-1.8
-1.8
-27.9
-5.9
xG diff
Possession diff
Shots on target diff
3 Results inside each profile
Dominating (n=44)
Equilibrated, best (n=60)
Equilibrated, worst (n=60)
Dominated (n=44)
82%
16%
55%
28%
17%
17%
28%
55%
16%
82%
Share of results
Win
Draw
Loss

THANK YOU FOR YOUR ATTENTION